VIVEK KUMAR MISHRA

EB-1A Extraordinary Ability Profile Summary

Applied Artificial Intelligence & Machine Learning in Critical Infrastructure

OVERVIEW

Demonstrated extraordinary ability in the sciences (Applied AI and Machine Learning), satisfying **6 out of 10** regulatory criteria under 8 C.F.R. § 204.5(h)(3). Evidence establishes sustained national and international acclaim, with methodologies permanently archived by the U.S. federal government and deployed at the highest levels of enterprise finance and public infrastructure.

CRITERIA MET (6 of 10)

1. Judging the Work of Others (8 C.F.R. § 204.5(h)(3)(iv))

- **Premier ML Conference Ecosystems:** Selected to evaluate scholarly work within the research ecosystems of the "Big Three" AI/ML conferences: **ICLR 2025** FM-Wild Workshop (1 paper reviewed), **NeurIPS 2025** Position Paper Track (3 papers reviewed), and **ICML 2026** AIWILD Workshop (2 papers reviewed).
- **IEEE Panelist:** Serves on the elite IEEE Senior Member Review Panel to evaluate the technical qualifications of peers globally.

2. Critical or Leading Role (8 C.F.R. § 204.5(h)(3)(viii))

- **JPMorgan Chase (#1 U.S. Bank):** Serving as Founding Engineer for the Chase Shopping Hub, a mission-critical platform for 80 million digital customers.
- **Verifiable Impact:** Designed architecture processing 5M+ monthly requests with 99.9% reliability; optimized backend AWS operations resulting in significant measurable enterprise cost savings ($100k+).

3. Original Contributions of Major Significance (8 C.F.R. § 204.5(h)(3)(v))

- **Institutional Book Adoptions:** Authored *Design Patterns for AI Agents*, formally integrated into the curriculum/libraries of three independent AICTE-regulated engineering institutions (Anand College, Supaul College, LNCTS Bhopal).
- **Federal & State Adoption:** Developed Python-based computational frameworks for transportation systems under USDOT/Caltrans funding; methodologies subsequently adopted across state lines by researchers at the **Florida Department of Transportation**.

4. Authorship of Scholarly Articles (8 C.F.R. § 204.5(h)(3)(vi))

- **Federal Archival:** Two publications permanently indexed in the U.S. **National Transportation Library** (ROSA P), establishing the work as authoritative federal technical knowledge.
- **Publication Record:** 9 peer-reviewed scholarly articles in applied AI and infrastructure optimization.
- **Citation Impact:** 48+ citations (Google Scholar, h-index: 4, accelerating trend with 22 citations in 2025) and 23 citations on ResearchGate; cited internationally by researchers in the U.S., Italy, China, and Iran.

5. Membership in Outstanding Associations (8 C.F.R. § 204.5(h)(3)(ii))

- **Fellow of BCS (FBCS):** Elected as a Fellow of the British Computer Society (The Chartered Institute for IT), the highest grade of membership reserved for elite professionals demonstrating outstanding IT leadership and major industry impact.
- **IEEE Senior Member:** Elevated to Senior Member grade, a highly selective distinction requiring 10 years of professional practice and 5 years of "significant performance," held by only the top 8% of the world's 460,000+ IEEE members.

6. Published Material in Major Media (8 C.F.R. § 204.5(h)(3)(iii))

- **Media Coverage:** Featured in *VoyageLA* ("Rising Stars"), *TechBullion*, *TechTarget*, and *VKTR*. Profiled for technical achievements in major Indian regional dailies (*Dainik Bhaskar*, *Rajasthan Patrika*).